

Evidence-Bounded Activity Discovery in London

A provenance-aware evaluation of OpenActive reconstruction and a diagnostic assurance instrument

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Decision sought. Approve one evidence-bounded discovery project, not a personalised recommender. It will determine what prespecified London finder and referral searches can be supported by observed OpenActive listings, identify why answers remain unresolved, measure the contribution of standards-aware reconstruction, and deliver a working Discovery Assurance Finder with publisher and pathway recommendations.

1. London Sport challenge, public value and scope

London Sport asks how innovative use of OpenActive data can make it easier for people across London's diverse communities to find and participate in physical activity. Its publishing, finder and referral pathways depend on information distributed across changing feeds, related event records and recurrence rules. A consumer may therefore suppress plausible listings or present unsupported certainty when acquisition fails, a required field is absent, a parent is unresolved, a Schedule is not expanded, a vocabulary does not map, or a downstream projection loses evidence.

Public-value pathway. This study assures the shared information layer upstream of public-facing finders and referral workflows. Better diagnosis enables London Sport, aggregators and publishers to distinguish 'no observed match' from 'insufficient evidence', improve source fields and consumer processing, and communicate uncertainty rather than fabricate certainty. The prototype is positioned at the analyst/referral-practitioner decision surface; effects on attendance and physical activity remain future evaluation.

Aim. To construct a reproducible London-serving OpenActive corpus; determine which prespecified discovery tasks it can answer; attribute indeterminacy to collection, publication, inheritance, schedule expansion, vocabulary, geography or consumer projection; and evaluate evidence policies and bounded interventions across frozen vintages.

Boundary. The unit of claim is an observed listing at a named snapshot, not London provision. 'No observed listed match' never means 'no activity exists'. Personalised recommendation, relevance ranking, clinical safety, causal health effects and population benefit are excluded. Capacity or real-time availability is reported only where explicitly published and is not inferred from recurrence. Diversity is operationalised through barrier-informed price, time, age, accessibility-information and location scenarios plus deprivation-stratified origins; ethnicity, language, cultural relevance, gendered preferences, trust and perceived safety are acknowledged but not inferred from fields that do not encode them. R3 is operationalised as the data-side answerability of Londoner-relevant searches, not observed user behaviour; R4 addresses equity-relevant catalogue answerability and bounded interventions, not measured reach, engagement or lived access. These interpretations require supervisor approval.

2. Research questions and contribution

Question	Research focus
RQ1	Which London-serving feeds, current records, inheritance chains, schedules and discovery fields can be reconstructed reproducibly, and what acquisition, semantic and spatial failures remain?
RQ2	For a frozen London benchmark, which searches yield supported matches, indeterminate candidates, bounded observed non-matches or scope-indeterminate records, and how do closed, permissive and explicit three-state policies trade false suppression against false assurance?
RQ3	How much benchmark-query determinability is carried directly, through inheritance and through Schedule expansion; how do documented consumer projections alter it; and which information interventions remain useful on later vintages?

Contribution and novelty. Breadth is methodological but the research object remains singular: exact-byte data engineering, temporal and geospatial reconstruction, bounded formal semantics, information-retrieval benchmark design, clustered statistical evaluation, research-grade software testing and partner implementation all answer the same discovery-assurance question. Generic mapping, coverage and field-completeness reporting are already occupied by the live public OpenActive Data Intelligence Platform (verified 18 July 2026; formal production/beta status unverified). Novelty is therefore a candidate, not an assertion: task-conditioned evidence-bounded discovery, mechanism-typed provenance, structural-informativeness decomposition and a consumer-processing conformance candidate. A primary-source capability audit must classify each capability as full, partial, absent or unclear across OpenActive tooling, deployed finders, aggregators and Active Partnerships before any novelty claim is frozen.

3. Corpus, reconstruction and evidence semantics

Each H1/H2/H3 snapshot retains original response bytes before parsing, requested and final URL, HTTP status and relevant headers, UTC retrieval time, retries and SHA-256. A versioned ledger records catalogue scope, selected/excluded feeds, publisher/platform evidence, feed relationships, licences, page acquisition and live-edge status. A bitemporal store separates observation/retrieval time from activity-valid time; updates and deletion tombstones materialise the current state at each vintage. Failed acquisition remains a collection-level coverage limitation; it never manufactures synthetic unknown records.

Stage	Representation	Research role
S0	Current source-published records; links retained; no explicit inheritance or recurrence expansion.	Direct-record ablation.
S1	EventSeries to SessionSeries to ScheduledSession inheritance; child overrides; full field lineage.	Inheritance contribution.
S1b	S1 plus valid Schedule expansion over the frozen horizon; PartialSchedule never expanded.	Primary reconstructed stage.
S2a	Versioned reconstruction of the historical 15-column pipeline, only if mapping provenance passes.	Bounded real-consumer case.
S2b	Pre-frozen synthetic lossy tabular projection.	Sensitivity stress test only.

Mechanism-typed evidence provenance

Every required atomic predicate stores T (support only), F (contradiction only), U_m (neither, with a closed unknown-mechanism label) or B_m (conflicting support and contradiction), and carries a versioned provenance graph linking result, transformation rule and source bytes. The public interface remains three-decision: a supported match requires every predicate to be T; a bounded known non-match requires at least one unconflicted F; U_m and B_m are conservatively exposed as indeterminate, with conflict distinguished from missing evidence. Mechanisms include source absence, linked-source failure, unresolved inheritance, invalid/ambiguous or conflicting values, vocabulary failure, projection loss, staleness and spatial uncertainty. Query answers retain the observed candidate universe and an independent acquisition-coverage qualifier.

Evidence grades. Schedule-derived occurrences are valid consumer reconstructions under the OpenActive specification, but are visibly graded as determinate-schedule-derived rather than determinate-explicit. Neither grade implies capacity, bookability, cancellation status or attendance without published evidence. Every generated occurrence traces to a full Schedule, rule version and deterministic identity; PartialSchedule is never extrapolated.

Formal boundary. A prespecified possible-worlds analysis is restricted to value completions of the fixed observed candidate universe. Proof obligations test both candidate-match soundness and bounded non-match soundness and, where separability and joint-realizability assumptions hold, exactness for a declared positive conjunctive fragment. Counterexamples characterise failures caused by conflicting or correlated predicates, repeated unknowns, negation and collection incompleteness. These are proof obligations, not assumed results; if exactness fails, the operational evidence model remains and the theoretical claim narrows. Acquisition gaps remain coverage qualifications outside record-value completion.

Executable validity controls

Before H1, the catalogue/feed iteration and optional-field drop defects are corrected and regression-tested. The corpus gate is an executable contract with per-snapshot metrics for attempted acquisition, current-state convergence, identifiers, lineage, Schedule handling, geocoding route, licence status and unexplained loss. Golden specification fixtures, an independently written reference interpreter, metamorphic tests and stateful property-based RPDE histories test DST, exceptions, explicit overrides, tombstones, replay idempotence and failure states. Synthetic corpora with planted mechanisms validate the full reconstruction-analysis pipeline and compare interval coverage under plausible dependence structures. Mutation testing of the evidence kernel reports killed mutants and classifies survivors as test gaps, specification ambiguities or equivalent mutants before the locked benchmark is evaluated.

4. London Discovery Benchmark and analysis

Family	Frozen task	Required evidence
Activity + geography	Named activity within a frozen distance	Mapped activity; validated London/distance evidence
Time / Schedule	Evening or weekend opportunity in the horizon	Explicit occurrence or valid full Schedule
Price visibility	Free or below a frozen price	Structured offer/price; absence remains unknown
Age information	Includes a stated age	Structured age range
Accessibility information	Publishes a requested access feature	Structured accessibility evidence
Compound referral	Time AND price AND accessibility constraints	All component evidence plus coverage qualifier

Locked instrument and adjudication

At least 30 prespecified population-weighted LSOA origins, at least 10 per London-specific deprivation tertile, are crossed with six families to give at least 180 locked queries. Development queries are permanently excluded. A development-only simulation jointly chooses the final origin count and adjudicated pair-sample size against prespecified confidence-width and assessor-hours constraints using the actual paired estimator and dependence scenarios, never locked effects. Query text, thresholds, fields, origins, geocoding cascade, codebook, seeds, prior sight and analysis code are frozen in a tagged commit.

All stable identities in the frozen London-serving corpus form the auditable candidate universe; there is no ranking or hidden top-k. Geography is classified as known-inside, known-outside or scope-indeterminate and dual-route coordinate/address disagreements are audited on a stratified validation sample. A probability sample of query-record pairs over-represents stage/policy disagreements while retaining non-zero inclusion probability for agreements. Two masked assessors independently reconstruct predicate evidence from source lineage using the codebook; a third arbitrates and audits 10% of agreements. Weighted reference labels estimate semantic-classification error and support policy comparisons and sensitivity analyses; they do not silently redefine unsampled outputs.

Estimands, uncertainty and falsification

For query q and stage s , $A(q,s)=1$ when the frozen evidence engine returns at least one supported match. The primary estimand is $\Delta = \text{mean}[A(q,S1b)-A(q,S0)]$, with exact paired transitions and a 95% interval. $S0$ to $S1$ and $S1$ to $S1b$ decompose the benchmark-query determinability carried by inheritance and recurrence structure. Simulation selects a cluster-respecting resampling method with acceptable coverage for the declared query-generation design. Human reference labels quantify engine error and bound result sensitivity. Publisher/platform leave-one-out analyses and $H2/H3$ provide ecosystem sensitivity and later-vintage persistence, not universal or out-of-sample claims.

Secondary outcomes are pre-enumerated and estimation-led: explicit versus schedule-derived evidence, family effects, result depth, unresolved mechanisms, projection effects and deprivation-stratified descriptions. Closed, permissive and explicit three-state policies are compared through false-suppression and false-assurance rates and a cost-ratio threshold curve; no unmeasured practitioner or resident cost is asserted. Equivalence testing is used only if a workflow-derived margin and adequate precision are frozen. Diagnostic positive/negative controls are excluded from the primary score.

Interventions. Publisher priorities are blocker and bounded-intervention analyses, not claims about realised repairs: missing values are unknown, so hypothetical field completion cannot reveal which value a publisher would supply. Best/worst-case bounds and minimal blocker sets are reported; actual later-vintage field resolution is analysed separately. Exact Shapley allocation of simulated determinability gain is optional only if predeclared interventions interact materially and is never described as causal responsibility.

5. Proof of concept, London Sport value and assurance

The London Discovery Assurance Finder is a diagnostic instrument for a London Sport analyst or community-referral practitioner, not a clinical or personalised recommender. For each structured search it exposes supported, indeterminate, bounded non-match and scope-indeterminate states; explicit versus schedule-derived evidence; predicate lineage; why-not blockers; stage/policy comparisons; feed coverage and freshness; and versioned CSV/JSON exports. This creates an assurance layer for existing finder, publishing and referral pathways without claiming that the instrument itself changes participation.

London Sport requirement	Research evidence and usable deliverable
R1 Understand OpenActive	Frozen catalogue/feed audit; exact-byte coverage; current state; lineage; Schedule use; licences; platform and failure metrics.
R2 Approaches across the country	Primary-source capability matrix covering the live Data Intelligence Platform, other OpenActive tools, deployed finders, aggregators and Active Partnerships; dated and bounded novelty claims.
R3 Discovery challenges	Locked London benchmark of data-side answerability; evidence states; coverage qualifiers; policy trade-offs and mechanism attribution - not claims about observed user behaviour.
R4 Model, prototype, recommendations	Mechanism-typed evidence model; working Finder; bounded publisher priorities and interface guidance; equity-relevant catalogue answerability only - no measured reach, engagement or lived access.
R5 Embed digital pathways	Actor/workflow map for finder and referral use; versioned interface boundary; refresh, escalation, indicators, risks and proposed 90-day pilot.

Industry-grade and research-grade acceptance

A pinned container, versioned schemas, structured logs, explicit failure states, data dictionary, runbook, licence/attribution register and retention/access policy support one-command reproduction. All suites are pytest-collectable and CI runs the same environment. Acceptance requires unit, golden, differential, property, metamorphic, stateful, mutation, integration, regression and failure-injection tests; a clean non-author reproduction of every headline table/figure; and a mechanised claim audit linking each conclusion to snapshot, denominator, code and permitted language. The benchmark and analysis plan are prospectively registered under embargo with prior sight declared; ACM artefact criteria and an RO-Crate research bundle are internal acceptance targets, not unearned badges or public-release claims.

Team and gates. The work is organised as four intellectually substantive, cross-reviewed streams: acquisition/current state; reconstruction/evidence semantics; benchmark/statistics/adjudication; and Finder/national audit/pathway integration. Before H1 each stream has a named lead and non-author reviewer in the private repository; every member authors code and report evidence, reviews another stream and can defend the integrated design. Quality gates are: ratification; corpus; semantic/fixtures; benchmark freeze; locked evaluation; later-vintage sensitivity; independent reproduction; claim and partner handover. The final deadline is 18 September 2026.

Ethics. The applicable PGT ethics route is confirmed before practitioner feedback is collected as research evidence. With approval, feedback tests whether determinate answers are actionable and indeterminate explanations intelligible; without approval it remains supervision and is excluded from analysis. Free text is hidden by default, raw access is controlled, and redaction is not treated as proof of anonymity.

Supervisor decisions requested

Please confirm: (1) whether the evaluated diagnostic Finder satisfies the proof-of-concept requirement; (2) the applicable PGT ethics route for practitioner construct validation; (3) whether the evidence-bounded claim boundary, including the declared R3/R4 interpretations, and R1-R5 mapping fit London Sport 1; and (4) approval to freeze the proposal and three operational protocols after named team ownership is recorded.

Selected foundations

Belnap (1977), A useful four-valued logic. Imielinski & Lipski (1984), Incomplete information in relational databases, JACM 31(4). Green, Karvounarakis & Tannen (2007), Provenance semirings, PODS. Fan & Geerts (2010), Relative information completeness. Razniewski & Nutt (2014), Adding completeness information to query answers over spatial databases, ACM SIGSPATIAL. Libkin (2016), Certain answers as objects and knowledge, Artificial Intelligence 232. Sakai (2016), Topic set size design, Information Retrieval Journal 19. Darari et al. (2020), Completeness and soundness guarantees for conjunctive SPARQL queries, Semantic Web 11(3). Console et al. (2020), Coping with incomplete data: recent advances, PODS. W3C (2013), PROV-O. OpenActive (accessed 18 July 2026), RPDE, feed relationships, harvesting, Schedules, Validator, Data Quality Explorer and Data Intelligence Platform. London Sport (2026), London Sport 1 brief and finder/referral materials. Sport England (2026), Inequalities Metric. University of Bristol (2026), SEMTM0044 unit specification.